

Inverse maps on open sets with global metric bounds

MathlibAnnex.BilipschitzOrientation.BilipschitzOpenData

structure

Records the maps, their inverse relations, and the quantitative bounds used in the orientation argument.

Statement

Fix $n \in \mathbb{N}_0$ and put $E = \mathbb{R}^n$ with the sup norm $\|\cdot\|_\infty$. The structure specifies open sets $S, T \subseteq E$ and maps $f, g : E \rightarrow E$. Their restrictions $f|_S : S \rightarrow T$ and $g|_T : T \rightarrow S$ are inverse bijections. Both maps have global Lipschitz bounds, and f has a positive global lower bound.

Definition

Choose S, T, f, g as above, constants $C_f, C_g \geq 0$, and a real number $\ell > 0$. Require the image and inverse relations

$$\begin{aligned} f(S) &\subseteq T, & g(T) &\subseteq S, \\ g(f(x)) &= x \quad (x \in S), & f(g(y)) &= y \quad (y \in T), \end{aligned}$$

and the three metric estimates

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \leq C_f \|x - x'\|_\infty \quad (x, x' \in E),$$

$$\|g(y) - g(y')\|_\infty \leq C_g \|y - y'\|_\infty \quad (y, y' \in E).$$

Thus the upper and lower estimates concern all ambient points. They are additional global requirements; the inverse relations themselves are imposed only on the two open sets.

Assumptions

The dimension n may be zero. The sets S and T are open; neither connectedness, nonemptiness nor finite measure is required. The maps f and g are defined on all of E .

Conclusion

The specified restrictions are inverse bijections between S and T , with the displayed global bounds available for subsequent arguments. The definition does not require f and g to be inverse at every point outside those sets.

These are chosen objects together with their stated properties, not an orientation or volume theorem. In particular, the global injectivity of f is a consequence of the lower estimate, rather than an additional stored condition.

Main citations

- Exact declaration and its source — `MathlibAnnex.BilipschitzOrientation.BiLipschitzOpenData`

Lean source signature (exact)

```
structure BiLipschitzOpenData (n : ℕ) where
  source : Set (Fin n → ℝ)
  target : Set (Fin n → ℝ)
  isOpen_source : IsOpen source
  isOpen_target : IsOpen target
  f : (Fin n → ℝ) → (Fin n → ℝ)
  g : (Fin n → ℝ) → (Fin n → ℝ)
  mapsTo_f : MapsTo f source target
  mapsTo_g : MapsTo g target source
  left_inv : Set.LeftInvOn g f source
  right_inv : Set.RightInvOn g f target
  fConstant : ℝ≥0
  lipschitzWith_f : LipschitzWith fConstant f
  gConstant : ℝ≥0
  lipschitzWith_g : LipschitzWith gConstant g
  lower : ℝ
  lower_pos : 0 < lower
  anti : ∀ x y, lower * ‖x - y‖ ≤ ‖f x - f y‖
```

In the source	Mathematical meaning
<code>(n : ℕ)</code>	The dimension of $E = \mathbb{R}^n$ with its coordinate sup norm; $n = 0$ is allowed.
<code>source : Set (Fin n → ℝ)</code>	The source set $S \subseteq E$.
<code>target : Set (Fin n → ℝ)</code>	The target set $T \subseteq E$.
<code>isOpen_source : IsOpen source</code>	The condition that S is open.
<code>isOpen_target : IsOpen target</code>	The condition that T is open.
<code>f : (Fin n → ℝ) → (Fin n → ℝ)</code>	The map $f : E \rightarrow E$, defined on the whole space.
<code>g : (Fin n → ℝ) → (Fin n → ℝ)</code>	The map $g : E \rightarrow E$, likewise defined on the whole space.
<code>mapsTo_f : MapsTo f source target</code>	$f(S) \subseteq T$.
<code>mapsTo_g : MapsTo g target source</code>	$g(T) \subseteq S$.
<code>left_inv : Set.LeftInvOn g f source</code>	$g(f(x)) = x$ for every $x \in S$.

In the source	Mathematical meaning
<code>right_inv : Set.RightInvOn g f target</code>	$f(g(y)) = y$ for every $y \in T$. These two inverse relations are restricted to the sets.
<code>fConstant : ℝ≥0</code>	The nonnegative constant C_f .
<code>lipschitzWith_f : LipschitzWith fConstant f</code>	The global bound $\ f(x) - f(x')\ _\infty \leq C_f \ x - x'\ _\infty$ for all ambient x, x' .
<code>gConstant : ℝ≥0</code>	The nonnegative constant C_g .
<code>lipschitzWith_g : LipschitzWith gConstant g</code>	The global bound $\ g(y) - g(y')\ _\infty \leq C_g \ y - y'\ _\infty$ for all ambient y, y' .
<code>lower : ℝ</code>	The real lower constant ℓ .

In the source	Mathematical meaning
<code>lower_pos : 0 < lower</code>	The strict positivity $\ell > 0$, unlike the merely nonnegative upper constants.
<code>anti : ∀ x y, lower * ‖x - y‖ ≤ ‖f x - f y‖</code>	The global lower bound $\ell \ x - y\ _\infty \leq \ f(x) - f(y)\ _\infty$ for every $x, y \in E$. This completes all 17 fields, without connectedness, nonemptiness or finite volume.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 23–29:

```
noncomputable section

open Set MeasureTheory Filter
open scoped ENNReal NNReal Topology

namespace MathlibAnnex
namespace BilipschitzOrientation
```

▼ Exact content identity

Declaration: `MathlibAnnex.BilipschitzOrientation.BilipschitzOpenData`

Accepted content SHA-256: 2cb8ff50fae1852b9922b699e50793a4a370c98a0b8828ddf16612476ea59dd1

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